

Module 3 -Test I Biochemistry

Question 1

Which enzyme regulatory mechanism does not cause conformational change?

- a. allosteric activation and inhibition;
- b. phosphorylation or other covalent modification;
- c. proteolytic cleavage
- d. protein protein interactions between regulatory and catalytic subunits, or between two proteins.
- e. None of the above

Question 2

Which one of the following is true?

- a. Equilibrium constants are additive.
- b. Thermodynamically favourable reactions have positive ΔG .
- c. A thermodynamically favorable reaction can be achieved by coupling it to a highly exergonic reaction through a common intermediate.
- d. Standard free energy change is multiplicative.
- e. Endergonic reactions are thermodynamically favourable.

Question 3

Choose the incorrect statement about 2,3-bisphosphoglycerate?

- a. It facilitates oxygen delivery to tissues
- b. It is formed from intermediate of glycolysis
- c. It is formed in cells of different tissues
- d. It is exclusively found in RBCs
- e. Its formation prevents net ATP formation

Question 4

Which one is not of practical importance of enzymes?

- a. Measurements of the activities of enzymes in blood plasma, erythrocytes, or tissue samples are important in diagnosing certain illnesses.

b. Enzymes are potential inheritable genetic disorders indicators. c. Many drugs act through interactions with enzymes. d. Some disease conditions may be caused by excessive activity of an enzyme

e. Enzymes are not important practical tools in chemical engineering, food technology, and agriculture.

Question 5

All of the following electron carriers are components of the mitochondrial electron transport chain except:

a.FAD

b.Co-enzyme Q

c. NAD⁺

d.NADP⁺

e. FMN

Question 6

Which of the following statements best characterizes ATP synthase?

a. oligomycin binds to ATP synthase, directly preventing ATP export

b.its catalytic function is to synthesize ATP in a reaction driven by a chemiosmotic potential

c it couples ATP export from the mitochondrial matrix to ATP synthesis

d. the low H⁺ ion concentration outside mitochondria establishes an electrochemical gradient that drives ATP synthesis

e.it is soluble protein found in the mitochondrial matrix

Question 7

Which of the following is incorrect about fructose

a. Free fructose can be obtained from dietary sources

b. In liver, it is metabolized faster than glucose

c. Muscle hexokinase phosphorylates it into Fructose-1-phosphate

d. Liver fructokinase is responsible for phosphorylation

e. It is common constituent of sweet diets

Question 8

During the fed state insulin

- a. Activates cAMP cascade
- b. Activates protein kinase A
- c. Activates glycogen phosphorylase
- d. Inhibit phosphodiesterase
- e. Activates glycogen synthase

Question 9

Which of the following processes produces the chemical gradient that drives the generation of ATP from ADP?

- a. electron transport ejects H^+ into the intermembrane space
- b. osmotic forces cause H^+ to bind with FAD in the mitochondrial matrix
- c. $FADH_2$ is regenerated from FAD
- d. NADH is regenerated from NAD^+
- e. electron transport causes H^+ to penetrate the inner mitochondrial membrane

Question 10

Which process makes muscle phosphorylase kinase fully active?

- a. Ca^{2+} binding
- b. Phosphorylation
- c. Glucose binding
- d. ADP binding
- e. A and B

Question 11

Serum enzymes that are least significant in the diagnosis of myocardial infarction

a.

Aspartyl transaminase (AST)

b.

Alkaline phosphatase (ALP)

c.

Creatine kinase (CK)

d.

Alanine transaminase (ALT)

e.

Lactate dehydrogenase (LDH)

Question 12

Which of the following is not related to Wernicke Korsakoff syndrome a.

Alcohol abuse

b.

Thiamine deficiency

c.

Increased transaldolase activity

d.

Decreased transketolase activity

e.

Abnormal eye movement

Question 13

Choose the incorrect statement about glucuronic acid

a.

It is the product of glucose oxidation

b.

It is a component of glycosaminoglycans

c.

It is sugar acid

d.

It can be formed from ascorbate

e.

It is involved in detoxification process

Question 14

Which one is correct about addition of an uncoupling agent to respiring mitochondria

a.

formation of ATP continues, but O₂ consumption ceases

b.

activity of the TCA cycle ceases

c.

NADH oxidation ceases

d.

ATP formation ceases, but O₂ consumption continues

e.

fatty acid oxidation ceases

Question 15

One of the following is correct about glycogen phosphorylase.

a.

TPP is a coenzyme required for its activity

b.

Glucose-6-phosphate is the direct product its activity

c.

It is an enzyme that synthesizes glycogen

d.

It acts on reducing ends of glycogen

e.

Limit dextrin stops its action

Question 16

Which of the following is correct about GLUT-4?

a.

It facilitates glucose uptake by intestinal cells

b.

Its membrane expression is activated by insulin

c.

It facilitates glucose uptake by brain cells

d.

Hypoglycemia facilitates expression of GLUT-4 on membrane of cells.

e.

It is found on membranes of red blood cells

Question 17

In the presence of rotenone:

a.

Cytochrome a3 is reduced

b.

Cytochrome c is reduced

c.

NADH is oxidized by electron transport,

d.

Cytochrome a is reduced by electron transport

e.

FADH₂ is oxidized by electron transport

Question 18

All of the following statements are true about gluconeogenesis except

a.

It is activated by insulin

b.

Requires transporter of intermediate across inner mitochondrial membrane

c.

It occurs during fasting condition

d.

Involves mitochondrial and cytosolic reactions

e.

It requires ATP

Question 19

Which of the following is not required for continuation of glycolysis towards formation of pyruvate?

a.

NAD⁺

b.

Inorganic phosphate

c.

Mg²⁺

d.

H₂O

e.

ATP

Question 20

Which one of the following statements is incorrect

a.

Formula milks are recommended for lactose intolerant individuals

The Dutch people are least susceptible for lactose intolerance

c.

Lactose intolerant people may encounter gastrointestinal problems

d.

Lactose intolerance is less common among Asians

e.

Lactase deficient individuals are advised to avoid lactose

Question 21

The inhibition that occurs between malonate ($\text{CO}_2^- \cdot \text{CH}_2\text{CO}_2^-$) and

Succinate ($\text{CO}_2^- \cdot \text{CH}_2\text{CH}_2\text{CO}_2^-$) for succinate dehydrogenase is a.

Substrate independent

b.

Competitive

c.

Uncompetitive

d.

Irreversible competition

e.

Non =E2=80=93 competitive and Uncompetitive

Question 22

Which one is wrong about allosteric inhibition

a.

V_{max} is reduced.

b.

Most allosteric enzymes do not possess a quaternary structure

c.

The inhibitor is not a substrate analogue.

d.

K_m is usually increased

e.

It is partially reversible when an excess substrate is added.

Question 23

Enzymes that catalyse group elimination to form double bonds are generally classified as

a.

Oxidoreductases

b.

Isomerases

c.

Ligases

d.

Lyases

e.

Transferases

Question 24

Which of the following is not likely precursor of gluconeogenesis?

a.

Glycerol

b.

Acetyl CoA

c.

Lactate

d.

Glucogenic amino acids

e.

Alanine

Question 25

Enzymes that have identical catalytic properties, but differ in their structure are

a.

Cofactors

b.

All enzymes are structurally similar

c.

Isoenzymes

d.

Zymogens

e.

Proenzymes

Question 26

Choose the incorrect statement regarding glucokinase

a.

It is not inhibited by glucose-6-phosphate

b.

It can switch between cytosol and nucleus

c.

It has broad specificity

d.

It becomes active after carbohydrate rich meal

e.

It is found in I-cells of pancreas

Question 27

Pyruvate dehydrogenase complex does not require

a.

TPP

b.

FAD

c.

Biotin

d.

NAD⁺

e.

Coenzyme A

Question 28

The rate limiting enzyme of glycolysis is

a.

Enolase

b.

Phosphoglucomutase

c.

Hexokinase

d.

Phosphofructokinase-1

e.

Aldolase

Question 29

Cyanide is poisonous as it stops respiration because of its:

a.

Inhibition of TCA cycle

b.

Combination with RBC cell membrane

c.

Inhibition of cytochrome oxidase

d.

Formation of complex with Hb

e.

Inhibition of myoglobin

Question 30

Choose the correct statement regarding muscle glycogenolysis a.

It results in release of free glucose to the blood

b.

It primarily occurs during fasting condition

c.

Epinephrine initiates muscle glycogenolysis during stressful situation d.

It occurs during fed state

e.

It is initiated in response to glucagon action

Question 31

The oxidation and phosphorylation in intact mitochondria is blocked by:

a.

Oligomycin

b.

Gentamicin

c.

Streptomycin

d.

Puromycin

e.

Septran

Question 32

Glycogen degradation is terminated by all of the following except a.

Hydrolysis of GTP bound to H-subunit

b.

Dephosphorylation of glycogen phosphorylase

c.

Degradation of glycogenin

d.

Degradation of cAMP

e.

Inactivation of adenylate cyclase

Question 33

Which one is not the salient features of non-competitive inhibition

a.

K_m is not changed

b.

K_m increase and V_{max} unchanged

c.

An inhibitor is not structurally analogous with a substrate

d.

V_{max} is reduced

e.

It is generally not reversible

Question 34

Which one of the following components of the electron transport chain only accepts electrons, and does not donate them?

a.

Coenzyme Q

b.

Cytochrome b

c.

FMN

d.

Cytochrome C1

e.

Oxygen

Question 35

The oxidative phase of pentose phosphate pathway has all of the following roles except

a.

Biosynthesis of nucleotides

b.

Biosynthesis of cholesterol

c.

Detoxification

d.

Biosynthesis of neurotransmitters

e.

Phosphorylation reactions

Question 36

Which of the following best explains how the brown adipose tissue in newborns generates large amounts of heat?

- a.
increased formation of ATP
- b.
increased availability of fat
- c.
increased availability of carnitine
- d.
mitochondrial leakage of hydrogen ions
- e.
decreased availability of carnitine

Question 37

All of the following statements describe PFK-1 except

- a.
It is allosteric enzyme
- b.
Its activity is increased in fed state
- c.

It catalyzes rate-limiting step of glycolysis

d.

Fructose-2,6-bisphosphate is potent inhibitor

e.

It is inhibited by high energy charge

Question 38

One of the following coenzymes is required for pentose phosphate pathway reactions

a.

FAD

b.

NAD⁺

c.

Biotin

d.

Coenzyme A

e.

TPP

Question 39

Which of the following is not true about glycolysis?

a.

It occurs in two phases

b.

Glycolysis can occur in the presence of oxygen

c.

The reactions take place in cytosol

d.

It provides precursors for biosynthesis of molecules

e.

There is no net ATP production during anaerobic glycolysis

Question 40

Electron transport chain carriers are located:

a.

In mitochondrial matrix

b.

In the mitochondrial intermembrane space

c.

On the inner surface of the outer mitochondrial membrane

d.

On the outer surface of the outer mitochondrial membrane

e.

In the inner mitochondrial membrane.

Question 41

Protein compound that do not participate in the catalytic process a.

Proenzymes

b.

Metal ions

c.

Coenzymes

d.

Metalloenzymes

e.

Cofactors

Question 42

Clinical important enzyme specific for diagnostic application of bone disease

a.

Alkaline phosphatase (ALP)

b.

Aspartate aminotransferases (AST)

c.

Gama-glutamyl transferase (GGT)

d.

Creatine kinase (CK)

e.

Alanine aminotransferases (ALT)

Question 43

Enzyme catalysis can be affected by all of the following except

a.

Substrate concentration

b.

Enzyme concentration

c.

Inhibitors or activators

d.

The catalytic efficiency of already produced enzyme

e.

Equilibrium constant

Question 44

One of the following pentose phosphate pathway enzyme is inhibited by increased NADPH

a.

Gluconolactonase

b.

Transketolase

c.

Glucose-6-phosphate dehydrogenase

d.

Transaldolase

e.

6-phosphogluconate dehydrogenase

Question 45

The free energy change for a reaction

a.

independent of the nature of the reaction.

b.

is dependent on the pathway by which the reaction occurs.

c.

does not affect by the nature and concentration of the reactants and products.

d.

Depends on the nature and concentration of the reactants and products.

e.

depends on only the concentration of the products.

Question 46

Which one is not true about the three thermodynamic quantities that describe the energy change occurring in a chemical reaction? a.

Enthalpy reflects the number and kind of chemical bonds in the reactants and products.

b.

AG expresses the amount of energy capable of doing work in a reaction at constant temperature and pressure.

c.

Change in enthalpy, temperature, and change in entropy never determine thermodynamic favourability.

d.

When the system possesses less free energy, the reaction is said to be exergonic.

e.

The reaction proceeds with a gain of entropy when the products of the reaction are less complex and more disordered than the reactants.

Question 47

One of the following glycosaminoglycans has anticoagulant role? a.

Dermatan sulfate

b.

Hyaluronic acid

c.

Heparin

d.

Chondroitin sulfate

e.

Keratin sulfate

Question 48

Which of the following enzyme is not common to both glycolysis and gluconeogenesis?

a.

Glucose-6-phosphatase

b.

Phosphoglycerate kinase

c.

Glyceraldehyde-3-phosphate dehydrogenase

d.

Aldolase A

e.

Enolase

Question 49

Which of the following tissues stores significant amount mobilizable glycogen in our body?

a.

Brain

b.

Liver

c.

Adipose

d.

Kidney

e.

RBCs

Question 50

The phosphate: Oxygen (P: O) ratio is defined as:

a.

The moles of ATP formed divided by the milligrams of protein

b.

The moles of CO₂ produced divided by moles of O₂ consumed c.

The moles of ATP synthesized divided by the atom equivalents of O₂ consumed

d.

The moles of phosphate consumed divided by moles of oxygen consumed

e.

The moles of O₂ consumed in presence of ADP divided by the moles of O₂ consumed in absence of ADP

Question 51

All of the following result from deficiency of Glucose-6-phosphate dehydrogenase except

a.

Lack of NADH

b.

Formation of Heinz bodies

c.

Hemolysis

d.

Increased oxidative stress

e.

Decreased sugar phosphates

Question 52

All are high-energy reactions except

a.

1,3-Bisphosphoglycerate⁴⁻ + H₂O → 3-phosphoglycerate³⁻ + Pi²⁻ + H⁺ (ΔG° = -49.3 KJ/mol)

b.

c.

Glucose 6-phosphate, with a ΔG° of hydrolysis of -13.8 kJ/mol

d.

Phosphocreatine²⁻ + H₂O → Creatine²⁻ + Pi²⁻ + H⁺ (ΔG° = -43.0 KJ/mol)

e.

Acetyl-CoA + H₂O → Acetate⁻ + CoA + H⁺ ($\Delta G^{\circ} = -31.5 \text{ kJ/mol}$)

e.

Hydrolysis of ATP ($\Delta G^{\circ} = -30.5 \text{ kJ/mol}$)

Question 53

A compound that serves as a rapidly mobilizable reserve of high-energy phosphates in skeletal muscle, myocardium and the brain

a. Phosphoenolpyruvate

b.

1,3-Bisphosphoglycerate

c.

Glucose-1-phosphate

d.

Thioesters

e.

Phosphocreatine

Question 54

Gluconeogenesis primarily occurs in

a.

Adipose

b.

Muscle

c.

Retina

d.

Liver

e.

Brain

Question 55

Choose the incorrect statement about pyruvate oxidation

a.

It takes place in RBCs

b.

In humans, it occurs in mitochondrial matrix

c.

It is oxidative decarboxylation reaction

d.

PDH complex consists of three enzymes

e.

It can occur in the presence of oxygen

Question 56

Two moles of R1 reduced by four moles of the biological electron carrier (R2) to produce three moles of product (P1) and oxidized electron carrier (P2). The mass action ratio will be

a.

$$([R1] [R2]) / ([P1] [P2])$$

b.

$$([P1]^3 [P2]) / ([R1]^3 [R2]^4)$$

c.

$$(3 [P1] [P2]) / (3 [R1] 4 [R2])$$

d.

$$(3 [R1] 4 [R2]) / (3 [P1] [P2])$$

e.

$$([R1]^3 [R2]^4) / ([P1]^3 [P2])$$

Question 57

One of the following is incorrect about dietary fiber?

a.

Prevents constipation

b.

Decreases serum cholesterol

c.

Decreases blood pressure

d.

Increases gastrointestinal diseases

e.

Promotes immune function

Question 58

Which one is an activation-transfer coenzyme

a.

Vitamin C

b.

Thiamine pyrophosphate

c.

NAD⁺

d.

FAD

e.

Vitamin E

Question 59

For hydrolysis reactions with large, negative ΔG° , the products are more stable than the reactants because:

a.

the bond strain in reactants due to electrostatic repulsion is relieved by charge separation.

b.

the products are stabilized by ionization.

c.

the products are stabilized by isomerization (tautomerization)

d.

the products are stabilized by resonance

e.

All are possible answers

Question 60

Identify a statement that is not the property of enzymes a.

Enzymes have an active site with amino acid residues.

b.

Accelerate the rate of reaction inducing high activation energy. c.

Enzymes are catalytically specific.

d.

Their catalytic activity depend on pH and temperature.

e.

Their catalytic activity depends on substrate concentration.